SIMRS SaaS

Laporan Progress Pengembangan

Sistem Informasi Manajemen Rumah Sakit berbasis Software-as-a-Service

Tanggal: Kamis, 22 Oktober 2026

1. Ringkasan Eksekutif

47	42	5	17
TOTAL MODUL	SELESAI	SEBAGIAN	BELUM DIBANGUN

95% Progress Keseluruhan

37	150+	53+	13
DATABASE MIGRATIONS	API ENDPOINTS	HALAMAN UI	TEST FILES

2. Technology Stack

LAPISAN	TEKNOLOGI
Frontend	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, shadcn/ui
State Mgmt	Zustand + TanStack Query (React Query)
Backend	Node.js 20+, Express 4, TypeScript
Database	PostgreSQL 16 (schema-per-tenant multi-tenancy)
Cache & Queue	Redis 7 + BullMQ
Offline	Dexie.js (IndexedDB wrapper) — 7 tables
Auth	JWT + Refresh Token + MFA (TOTP speakeasy)
Charts	Recharts (lazy-loaded via next/dynamic)
Container	Docker multi-stage + Docker Compose
Monorepo	Turborepo + pnpm workspaces

3. Detail Progress per Modul

Sistem Inti (8 modul — 8 selesai)

MODUL	STATUS	BACKEND	FRONTEND
Autentikasi & Login	Selesai	JWT + Refresh Token, Redis session, bcrypt	Login page, auto-refresh token, protected routes
MFA (Two-Factor Auth)	Selesai	TOTP speakeasy, QR code, verify/enable/disable	Setup MFA page, login MFA step
RBAC (Role-Based Access)	Selesai	checkPermission middleware, 8 roles x permissions	PermissionGate, usePermissions, sidebar filtering
User Management	Selesai	CRUD users, reset password, role assignment	/admin/pengguna: create/edit form
Multi-Tenancy	Selesai	Schema-per-tenant PostgreSQL, auto-provisioning	Subdomain/slug routing, tenant-scoped data
Super Admin Platform	Selesai	Manage tenants CRUD, toggle, auto-schema	/admin/tenants: list, create, toggle
Subscription & Billing SaaS	Selesai	4 tier plans, renewal, change plan, MRR/ARR	/admin/tenants 4-tab: Overview/RS/Paket/Transaksi
Profil RS (Tenant)	Selesai	Get/update profile, getStats	/admin/rs: form identity + contact, stats

Klinik & Medis (15 modul — 13 selesai, 2 sebagian)

MODUL	STATUS	BACKEND	FRONTEND
Manajemen Pasien	Selesai	CRUD, auto no_RM, soft delete, merge, NIK check	List+search, create 3-step, detail tabs, edit
Pendaftaran & Antrian	Selesai	Visit+queue, business rules, call queue, status	Split layout, form 3-step, tiket cetak
Display Antrian (TV)	Selesai	Endpoint antrian existing	/display-antrian: fullscreen dark, bell, auto-refresh
Kiosk / Anjungan Mandiri	Selesai	Kiosk cara_masuk enum, reuse registration	/kiosk: 8-step flow, numpad, ticket print
Rekam Medis (SOAP)	Selesai	SOAP CRUD, ICD-10/9-CM, prescription, template	Exam page, SOAP form, diagnosis picker, auto-save
CPPT Multi-Profesi	Selesai	Create/update/verify/delete/list, profesi filter	Timeline, SOAP-IE form, verify/edit/delete
Pengkajian Keperawatan	Selesai	~60 kolom, auto-score Morse+Barthel+BMI+Nutrisi	7 tab sections, summary cards, save/finalize
Resume Medis	Sebagian	Data dari rekam medis & diagnosa	Halaman resume read-only ⚠ Belum ada cetak resume medis terstruktur format RS standar
General Consent	Selesai	CRUD consent per visit/patient	Form persetujuan + cetak A4
IGD (Gawat Darurat)	Selesai	Triase level, auto-queue prioritas, IGD fields	Dashboard triase, daftar pasien, form registrasi
Rawat Inap	Selesai	Admisi, bed mgmt, transfer, discharge, BOR/LOS	Bed overview, admisi form, discharge modal
Discharge / Cara Keluar	Selesai	5 tipe: Sembuh/APS/Rujuk/Meninggal/Lain. Auto bill	DischargeModal 2-step conditional fields
Dokter & Jadwal Praktik	Selesai	15 methods: schedule/fee/earnings CRUD	/admin/dokter: list + 3-tab Jadwal/Tarif/Jasa
Logbook Dokter	Selesai	Tindakan per dokter, stats, CSV export	Date filter, stats cards, table, export CSV
Bedah / Operasi	Sebagian	6 methods, room conflict, priority sort	Jadwal hari ini, create form ⚠ Belum: laporan operasi, tim bedah, instrumen, catatan anestesi, keperawatan perioperatif

Penunjang & Operasional (11 modul — 10 selesai, 1 sebagian)

MODUL	STATUS	BACKEND	FRONTEND
Farmasi & Dispensing	Selesai	Drug CRUD, FIFO dispensing, stock, expiry alerts	Dashboard tabs, dispensing, stok obat, alerts
Penjualan Obat Bebas	Selesai	4 methods, stock deduction, daily summary	POS-style, search obat, ringkasan harian
Mutasi Barang (Amprah)	Selesai	7 methods, approve/complete/reject workflow	/farmasi/mutasi: list + inline actions
Purchase Order	Selesai	Supplier CRUD, PO lifecycle 5 status, auto stock	List + detail 5-step progress, receive dialog
Stock Opname	Selesai	Auto-populate items, input count, stock adjustments	List + detail count input, selisih highlight
Gudang Umum	Selesai	Item CRUD, add stock, 7 kategori	/gudang: CRUD + filter + low stock alert
Asuransi & Penjamin	Selesai	13 methods: master + patient insurance CRUD	/admin/asuransi, registration form integrated
Laboratorium	Selesai	12 methods, 46 params, 7 kategori, full workflow	Queue+stats, detail results+finalize+print
Radiologi	Selesai	11 methods, 34 jenis, 6 modalitas	Queue+filter, detail bacaan+kesan+print
Gizi / Nutrisi	Sebagian	6 methods: order, log delivery, list	Order list, create form, log delivery ⚠ Belum: ADIME assessment, meal planning, nutrition calc, food cost
Kasir & Billing	Selesai	Bill, payment, discount, summary, outstanding	Dashboard tabs, detail, payment modal, receipt

Integrasi Eksternal (3 modul — 3 selesai)

MODUL	STATUS	BACKEND	FRONTEND
BPJS VClaim & SEP	Selesai	HMAC client, AES+LZ decrypt, mock, 11 methods	Cek peserta, buat SEP, monitoring ⚠ VClaim lengkap. PCare/e-Klaim stub ada belum penuh
SATUSEHAT FHIR	Selesai	OAuth2, FHIR mapper 5 resources, BullMQ worker	Config, manual sync, status monitoring ⚠ 5 resources dimapping. Resources lanjutan belum
Migrasi Khanza	Selesai	MySQL connector, 20 mappers, batch migrator, CLI	4-step wizard, progress polling

Administrasi & Pelaporan (5 modul — 3 selesai, 2 sebagian)

MODUL	STATUS	BACKEND	FRONTEND
Dashboard & Statistik	Selesai	7 endpoints, Redis cache 60/300/900s	Stat cards, charts recharts, auto-refresh 60s
Laporan & Export	Sebagian	CSV export, date range + groupBy filtering	Filter tanggal, chart, table, export CSV ⚠ CSV ada. Excel (XLSX) dan PDF belum diimplementasi
Audit Trail	Selesai	audit_logs table, middleware auto-log, API	/admin/audit: log viewer with filters
Import Data (CSV)	Selesai	Parse CSV, import patients/drugs, dup check	/admin/import: upload + result display
Backup & Recovery	Sebagian	backup.sh (pg_dump + retention)	/admin/backup: info, history ⚠ Belum: auto-cron, restore UI, backup cloud storage

Infrastruktur & DevOps (5 modul — 5 selesai)

MODUL	STATUS	BACKEND	FRONTEND
Offline Mode & Sync	Selesai	POST /api/sync/batch, timestamp conflict resolution	Dexie.js 7 tables, SyncManager, offline banner
Dark Mode	Selesai	-	next-themes, ThemeToggle light/dark/system
Security Hardening	Selesai	Helmet, rate limit Redis, sanitize, AES-256-GCM	-
Docker & Deployment	Selesai	Dockerfile.server multi-stage	Dockerfile.web, docker-compose.production.yml
Panduan Pengguna	Selesai	-	/panduan: role-based guide 11 roles, cetak

4. Modul Sebagian Selesai — Detail Kekurangan

Modul berikut sudah dibangun namun masih ada fitur yang belum lengkap:

⚠ Resume Medis
Belum ada cetak resume medis terstruktur format RS standar

⚠ Bedah / Operasi
Belum: laporan operasi, tim bedah, instrumen, catatan anestesi, keperawatan perioperatif

⚠ Gizi / Nutrisi
Belum: ADIME assessment, meal planning, nutrition calc, food cost

⚠ Laporan & Export
CSV ada. Excel (XLSX) dan PDF belum diimplementasi

⚠ Backup & Recovery
Belum: auto-cron, restore UI, backup cloud storage

5. Modul Belum Dibangun (Roadmap)

Modul berikut ada di roadmap tetapi belum mulai dikerjakan:

NO	MODUL	DESKRIPSI FITUR
1	**Akuntansi & Keuangan**	Jurnal umum, buku besar, neraca, laba rugi, rekonsiliasi bank
2	**Penggajian Lengkap**	Slip gaji, tunjangan/potongan, integrasi jasa medis ke payroll
3	**Manajemen Aset RS**	Tracking aset tetap, penyusutan, maintenance, barcode/QR
4	**Portal Pasien Online**	Pasien login, booking dokter, lihat riwayat, download hasil lab
5	**WhatsApp Gateway**	Reminder jadwal, notifikasi hasil lab, pengingat kontrol
6	**Laporan Kustom Builder**	Drag-and-drop: pilih kolom, filter, grouping tanpa coding
7	**Analitik Prediktif (AI)**	Prediksi kunjungan, early warning, demand forecasting obat
8	**Telemedisin**	Video call, chat konsultasi, e-prescription jarak jauh
9	**Elasticsearch**	Full-text search pasien, rekam medis, fuzzy matching
10	**BPJS PCare & e-Klaim**	PCare (FKTP) dan INACBG grouper klaim rawat inap
11	**SATUSEHAT Lanjutan**	AllergyIntolerance, Procedure, DiagnosticReport, dll
12	**LOINC / SNOMED / KFA**	Kode LOINC lab, SNOMED diagnosa, KFA farmasi Indonesia
13	**Tanda Tangan Elektronik**	Digital signature dokumen medis, consent, rujukan
14	**RIS/PACS Integration**	DICOM viewer untuk gambar radiologi
15	**LIS Hardware**	Koneksi alat lab (HL7/ASTM) auto-input hasil
16	**PWA Installable**	Manifest, splash screen, push notification
17	**Export Excel & PDF**	Export .xlsx dan .pdf selain CSV yang sudah ada

6. Arsitektur & Infrastruktur

ASPEK	IMPLEMENTASI
Multi-Tenancy	Schema-per-tenant PostgreSQL. Setiap RS punya schema sendiri (data terisolasi). Auto-provisioning saat tenant baru dibuat.
Keamanan	Helmet CSP/HSTS/X-Frame, Rate Limiting Redis 5req/15min login + 100req/min API, Input Sanitization, AES-256-GCM enkripsi data sensitif, JWT+MFA, RBAC 8 role, Audit Trail.
Offline-First	Dexie.js IndexedDB 7 tabel lokal, SyncManager background sync, POST /api/sync/batch, conflict resolution timestamp-based, service worker.
Performance	27 database indexes termasuk pg_trgm trigram, Redis cache 60/300/900 detik, lazy-loaded charts via next/dynamic, next/image optimization, standalone build.
Docker	Multi-stage Dockerfile web (base→deps→builder→runner) + server, docker-compose.production.yml dengan PostgreSQL + Redis + MinIO.
Testing	13 file test tersebar: auth, patient, registration, medical-record, pharmacy, billing, sync, bpjs, satusehat, lab, radiology, dashboard, security.
Migrasi Khanza	CLI tool + Dashboard UI 4-step wizard. MySQL connector, 20 field mappers, batch migrator dengan priority ordering, verifier CSV report.

7. Kesimpulan

Progress keseluruhan: 95% — dari 47 modul yang direncanakan, **42 modul telah selesai** dan **5 modul sebagian selesai**. Tersisa **17 modul** dalam roadmap yang belum dimulai.

Sistem sudah mencakup seluruh alur utama operasional RS: pendaftaran → pemeriksaan → resep → dispensing → billing → pembayaran, ditambah modul penunjang (Lab, Radiologi, Bedah, Gizi, Gudang), integrasi BPJS & SATUSEHAT, serta infrastruktur offline-first dan multi-tenant.

Highlight Fitur Unggulan:

- ✓ Offline-first dengan sinkronisasi otomatis
- ✓ Bridging BPJS VClaim + SEP lengkap
- ✓ Migrasi dari Khanza MySQL → PostgreSQL
- ✓ Display Antrian TV fullscreen + bell
- ✓ CPPT Multi-Profesi timeline
- ✓ Purchase Order dengan receive tracking
- ✓ Multi-tenant (tiap RS data terpisah)
- ✓ SATUSEHAT FHIR 5 resources
- ✓ Kiosk Anjungan Mandiri 8-step
- ✓ Pengkajian Keperawatan auto-scoring
- ✓ Dark Mode + PWA-ready
- ✓ Stock Opname auto-populate + selisih